
SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I S. Praaveen (192524239) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

S. Praaveen
Signature of the student:

level -3

①

1) Delivery Drone Problem Solving Agent:

The most suitable agent is goal based problem solving agent.

Reason:

A delivery drone has a specific goal: deliver the package from the warehouse to the customer's location safely & efficiently. The agent must:

- ★ Analyse the environment
- ★ Plan the best route
- ★ Avoid obstacles and restricted areas
- ★ Reach the destination with minimum time and energy.

A goal based agent searches for the best sequence of action that satisfies the goal.

Problem Formulation:

- ★ State: Current drone location, battery level, package status.
- ★ Initial state :- Drone is at the warehouse with the package.
- ★ Goal state: Package delivered successfully to the customer's location.
- ★ Path cost: Distance travelled, battery usage and delay.

(2)

Working:

- ★ Drone starts at warehouse
- ★ Reviews delivery location
- ★ Calculate the shortest safe path.
- ★ Moves step by step
- ★ Reaches destination
- ★ Delivers package

Conclusion:

A Goal based problem solving agent is ideal because it plans actions based on the desired goal and selects the best route to deliver packages efficiently.

2) Iterative Deepening Search (IDS):

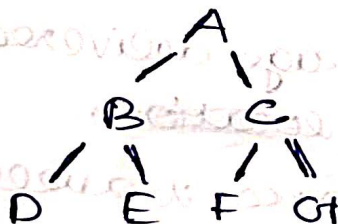
Definition:

Iterative Deepening Search (IDS) is a search algorithm that repeatedly performs Depth-first search (DFS) with increasing depth limits until the goal is found.

It combines:

- ★ Breadth first search (BFS) completeness
- ★ Depth first search (DFS) low memory usage.

Example:



Goal Node = F

Depth Limit = 0

search only root

A

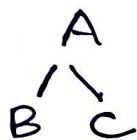
③

Visited order:

A

Goal Found? NO

Depth Limit = 1

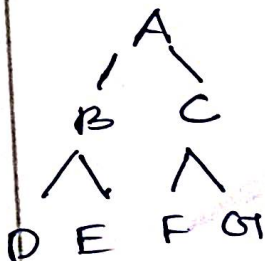


Visited order

A → B → C

Goal found? NO

Depth Limit = 2



Visited order

A
B
D
E
C
F

Goal found? YES

Advantages:

★ Suitable for large spaces.

★ Complete search

Disadvantages:

★ Repeats exploration

★ Slightly more computation than BFS

Conclusion:

IDS combines the space efficiency of DFS with the completeness of BFS, making it effective.

③ Missionaries & Cannibals Problem:

Problem Statement:

Three Missionaries (M) & three Cannibals (C) must cross a river using a boat that can carry at most two people.

Constraint:

On either bank,

Cannibals should never outnumber mission-

(4)

wherever missionaries are present.

Problem Formulation:

CM-Left, C-Left, Boat

Boat position:

• Left
• Right

Initial State:

(3, 3, Left)

Goal State:

(0, 0, Right)

Goal test:

Missionaries Left = 0

Cannibals Left = 0

Boat = Right

Breadth First Search:

State 1:

Left Bank:

Initial

MMM CCC

(3, 3, Left)

Boat

State 2:

Move 2:

Left Bank:

Right Bank:

(3, 1, Right)

MMM C

CC

Boat

State 3:

Left Bank:

Right Bank:

(3, 2, Left)

MMM CC

C

Boat

BFS Search order:

(3, 3, L)

(BFS continue

(0, 0, R)

↓

exploring all

(3, 1, R)

valid states

↓

level by

(3, 2, L)

searching)

Conclusion:

For finding the shortest solution it is most welcomed and useful.